

Five-Number Summary Box Plot

Unit 8: Sports Analytics Lab • Lesson 8-5 • EduWonderLab

UNIT 8 • LESSON 8-5 • TEACHER NOTES

Standard 6.SP.4

FORMAT Order data + build box plot

TIME 25–35 min

GROUPING Pairs

TOPIC Display Data: Box Plots

STANDARD 6.SP.4

PREP LEVEL Light — print data cards and box-plot template

Learning goal *I can find a five-number summary and use it to make a box plot.*

Teacher setup

- Print one card set and template per pair.
- Students order each data set and find the five-number summary.
- They draw the box plot; partners verify the quartiles.

Materials

Data cards, Box plot template, Five-number summary checklist, Pencil.

Differentiation & language support

- Checklist: minimum, Q1, median, Q3, maximum (SPED).
- Sentence stem: “The median is ___, and the quartiles are ___ and ___.”
- ESOL: order the numbers first, then mark the middle.

Key vocabulary (English / Español):

Term	Español	What it means
five-number summary	resumen de cinco números	min, Q1, median, Q3, max
quartile	cuartil	a value that divides data into quarters

Quick teacher check: *Check the median and quartiles on Card 2 (even number of values).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can find a five-number summary and use it to make a box plot.*

What you need

Data cards, Box plot template.

How to play

1. Order the data set from least to greatest.
2. Find the median, then Q1 and Q3.
3. Write the five-number summary.
4. Draw the box plot and have your partner check it.

Data Cards

Order the data, then find the minimum, Q1, median, Q3, and maximum.

1. 2, 4, 5, 7, 9, 10, 12	2. 1, 3, 3, 5, 7, 8
3. 10, 12, 15, 18, 20, 22, 25	4. 3, 5, 6, 8, 9, 11
5. 20, 24, 30, 36, 40, 44, 50	6. 4, 4, 6, 8, 10, 12, 14, 16

Record your work (five-number summary):

Explain your thinking

Explain what the box of a box plot shows about the middle of the data.

Sentence frame: *The median is ___, and the quartiles are ___ and ___.*

★ **Challenge:** For Card 1, find the interquartile range (IQR). ($Q3 - Q1 = 10 - 4 = 6$)

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 2, 4, 5, 7, 9, 10, 12 Answer: min 2, Q1 4, median 7, Q3 10, max 12	2. 1, 3, 3, 5, 7, 8 Answer: min 1, Q1 3, median 4, Q3 7, max 8
3. 10, 12, 15, 18, 20, 22, 25 Answer: min 10, Q1 12, median 18, Q3 22, max 25	4. 3, 5, 6, 8, 9, 11 Answer: min 3, Q1 5, median 7, Q3 9, max 11
5. 20, 24, 30, 36, 40, 44, 50 Answer: min 20, Q1 24, median 36, Q3 44, max 50	6. 4, 4, 6, 8, 10, 12, 14, 16 Answer: min 4, Q1 5, median 9, Q3 13, max 16

Teaching notes & look-fors

- Median splits the data in half; Q1 and Q3 are the medians of each half.
- Card 1 IQR = $10 - 4 = 6$.